Practical 01: Python & Machine Learning Environment Setup

Department of Computer Science Engineering — DBATU

Student Name: Ammara Sirkhot	**PRN:** 24030332905059
Department: Computer Science Engg.	**Course:** Machine Learning Lab
Faculty Guide: Prof. Harsha Ma'am	**Academic Year:** 2025–2026
University: Dr. Babasaheb Ambedkar Technological University (DBATU), Lonere	

1. Aim / Objective

To set up a local Python and Machine Learning development environment using Anaconda/Virtual Environments and install essential data science libraries including NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, and Jupyter Notebook.

2. Prerequisites & Tools Installed

- **Python 3.x / Anaconda Navigator**
- **Virtual Environment Tools:** venv / conda
- **Core ML Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow
- **Development Environment:** Jupyter Notebook / VS Code

3. Step-by-Step Execution Guide

Step 1: Create a Project Folder

```
mkdir ML_Lab_Practicals
cd ML_Lab_Practicals
```

Step 2: Create an Isolated Python Virtual Environment

```
# Option A: Using venv
python -m venv ml_env

# Option B: Using Conda
conda create --name ml_env python=3.10 -y
```

Step 3: Activate the Virtual Environment

```
# Windows (CMD):
ml_env\Scripts\activate

# Linux / macOS:
source ml_env/bin/activate

# Conda:
conda activate ml_env
```

Step 4: Install Core Machine Learning Libraries

```
pip install jupyter notebook numpy pandas matplotlib scikit-learn tensorflow
```

4. Environment Verification Script

```
import numpy as np
```

```
import pandas as pd
import matplotlib.pyplot as plt
import sklearn
import tensorflow as tf

print("--- Machine Learning Environment Verification ---")
print(f"NumPy Version: {np.__version__}")
print(f"Pandas Version: {pd.__version__}")
print(f"Scikit-learn Version: {sklearn.__version__}")
print(f"TensorFlow Version: {tf.__version__}")
print("Status: All ML Libraries installed successfully!")
```

5. Launching Jupyter Notebook

```
jupyter notebook
```

6. Learning Outcomes

- Successfully isolated development dependencies using virtual environments.
- Configured essential data analysis and machine learning frameworks (NumPy, Pandas, Scikit-learn, TensorFlow).
- Set up an interactive Jupyter Notebook workspace for ML experiments.